

呼吸声学、咳嗽、呼吸与语音健康数据集版图报告（扩展版）

生成日期：**2026-06-17**

项目语境：**acoustic-based respiratory disease detection**

扩展目标：扩大 **dataset** 范围；新增公开数据集论文、自采集但可复现/模型定义充分论文；方法地图至少 **40** 篇论文且至少 **10** 篇为近两年工作。

重要限定：本报告只讨论数据集、标签、协议与机器学习研究路线，不构成临床诊断、筛查或疗效判断。凡无法通过官方数据页、挑战页、Zenodo/PhysioNet/Mendeley/GitHub/Hugging Face、原始论文或机构页面核验的字段均标为 **unknown / not found / requires package inspection**。本报告显式区分疾病标签、症状标签、声学事件标签、自报告标签、PCR-linked 标签和专家/医师听诊标注。

1. Executive Summary

本扩展版共覆盖 **52** 个数据集/数据资源，其中 **Tier 1 = 6**，**Tier 2 = 20**，**Tier 3 = 16**，**Tier 4 = 10**。相比上一版，本版新增或显著扩展了 **VocalSound**、**Audio-IMU Multimodal Cough**、**UCL Speech Breath Monitoring / ComParE 2020 Breathing**、**ESC-50**、**cough-speech-sneeze**、**OpenSLR Nonverbal**

Vocalization、**Resp-229K**、**FABS**、**DeepBreath/Pneumoscope**、**Bridge2AI-Voice Adult/Pediatric**、**RALE**、**MEEI**、**AudioSet Cough class** 等资源。

方法地图共列出 **56** 篇论文/数据论文/可复现方法资源，其中 **23** 篇为 **2024-2026** 年近两年工作。这些条目被分为三类：第一类是使用公开或半公开呼吸/咳嗽/语音健康数据集的论文；第二类是自采集或私有数据论文，但满足至少一个可复现条件，即有公开 repo、公开数据访问路径、或论文中模型/协议定义足够明确；第三类是泛音频/健康音频 foundation model 或辅助数据集论文。不同论文的结果只有在 相同数据集、相同标签、相同 **split**、相同指标和相同 **subject-independence** 约束 下才被视为可比。

短期最强路线仍然是 **respiratory acoustic event detection + cross-dataset robustness**。建议主线使用 **ICBHI 2017**、**HF_Lung_V1**、**SPRSound** 做 lung/breath event modeling，使用 **COUGHVID**、**Audio-IMU Cough**、**Corp Dataset**、**Clinic Waiting Room Cough** 做 cough/event detection 与质量控制，使用 **UK COVID-19 Vocal Audio** 做更可靠的 PCR-linked disease-label 分析。COVID 音频路线应把 **UK COVID Vocal** 和 **Coppock et al. 2024** 的负结果作为核心基准，避免只报告音频模型 AUC 而不比较症状基线。

当前直接可行的 SOTA 方向集中在三类：第一，ICBHI 上的 respiratory sound classification，例如 **MVST 2024**、**BTS 2024**、**RepAugment 2024**、**Stethoscope-guided SupCon 2024**、**ADD-RSC 2025**、**Architecture-Agnostic KD 2025**、**Geometry-aware AST+SAM 2025**、**Resp-Agent 2026**；第二，OPERA/HeAR/respiratory pretraining，例如 **OPERA 2024**、**HeAR 2024**、**Niizumi et al. 2025 M2D+Resp**；第三，COVID/cough validity route，例如 **UK COVID Vocal 2024**、**Coppock et al. 2024**、**SympCoughNet 2025**、**COUGHVID U-Net/DL 2025**。

2. Search Strategy And Inclusion Criteria

2.1 搜索策略

本版检索以一手来源为优先：官方数据页、挑战官网、Zenodo、PhysioNet、Mendeley Data、Dryad、GitHub/GitLab、Hugging Face、论文主页、期刊/会议论文页面和机构页面。关键词包括：respiratory sound dataset, lung sound dataset, wheeze dataset, crackle dataset, adventitious lung sound, pediatric respiratory sound, cough dataset, cough event detection, COVID cough, PCR referenced vocal audio, breathing sound dataset, speech-breath monitoring, voice pathology dataset, voice biomarker dataset, health acoustic foundation model, respiratory acoustic foundation model。

2.2 纳入标准

数据集或论文至少满足一项：

1. 直接包含肺音、呼吸音、咳嗽、呼吸、wheeze、crackle、rhonchi、stridor、sniff、sneeze 等呼吸相关声学内容；
2. 包含 COVID、TB、COPD、asthma、pneumonia、bronchiolitis、voice pathology 等呼吸/声学健康相关标签，且音频模态与咳嗽、呼吸、肺音、语音或嗓音健康相关；
3. 虽非疾病数据，但对预训练、域适配、负样本构造、语音健康或远程健康监测有明确方法学价值；
4. 对自采集/私有数据论文，只有在 **repo** 公开、数据访问路径公开、或模型/协议定义足够明确 时纳入 paper map；
5. 对无法直接行动但领域常见的资源，仅列为 Tier 4，不作为核心实验依据。

2.3 排除标准

排除仅含非音频生理信号且与声学无直接连接的数据集、无法核验来源的网页列表、普通 ASR/语音识别数据集、只含教学示例且缺少研究标注的音频库、以及无法确认标签语义/访问路径的镜像数据。Kaggle/YouTube/Freesound 镜像只在可以追溯到原始数据源时作为辅助或 Tier 4 处理。

2.4 Relevance tiers

层级	定义	本项目用法
Tier 1	Strong direct fit	可作为核心训练、评测或初步结果主数据集。直接支持 respiratory event detection、cough/breath analysis、lung sound classification 或 PCR-linked disease audio modeling。
Tier 2	Useful with caveats	相关性强但存在自报告标签、受限访问、小规模、协议不一致、临床元数据不足、设备/人群 mismatch 等问题。适合补充实验、外部验证或域适配。
Tier 3	Auxiliary / background evidence	适合预训练、负样本、语音健康背景、foundation model baseline 或方法启发；不能单独支撑核心呼吸疾病检测结论。
Tier 4	Mention only / low actionability	访问不可控、标签/协议不明、教育库/镜像/非音频或弱连接资源。只用于说明潜在线索或不可行动性。

3. Dataset Coverage Table

#	数据集/资源	层级	访问状态	主要任务路线	音频模式	关键规模	标签粒度	源
1	ICBHI 2017 Respiratory Sound Database	T1	open / mirror	event detection; lung sound classification	chest respiratory/lung sounds	920 recordings; 6,898 annotated respiratory cycles; 126 subjects; about 5.5 h	cycle-level wheeze/crackle; subject/recording metadata	sr c
2	HF_Lung_V1	T1	open GitLab	event detection; breath phase; adventitious sounds	lung sounds from electronic stethoscope	9,765 15-s files; 34,095 inhalation, 18,349 exhalation, 13,883 CAS, 15,606 DAS labels	onset-offset event labels for I/E/CAS/DAS	sr c
3	SPRSound	T1	open GitHub / challenge releases	pediatric respiratory event detection	pediatric lung sounds	2022 train 1,949 and test 734 records; original paper reports 2,683 records, 292 participants, 9,089 events	record-level classes and event onset-offset annotations	sr c
4	COUGHVID	T1	open Zenodo / EPFL page	cough quality; cough segmentation; weak disease prediction	crowdsourced cough recordings	over 30k recordings; 25k+ in Scientific Data paper; 2,800+ expert-labeled coughs	file-level cough quality/type; self-report COVID/ symptom metadata	sr c
5	UK COVID-19 Vocal Audio / COVID-19 Vocal Audio	T1	open Zenodo for cough/ exhalation; speech excluded from open release	PCR-linked disease prediction; cough/breath analysis	cough, exhalation, speech collected; open release contains cough and exhalation	72,999 participants; 70,794 PCR-linked; 53.7 GB open release	PCR-linked subject/session labels; symptom metadata	sr c
6	Corp Dataset	T1	paper/page listed; access path unclear	cough event detection in long recordings	long-duration cough audio from respiratory-disease patients	168 h; 9,969 cough events; 42 respiratory-disease patients	cough onset-offset; disease at subject level	sr c
7	KAUH / Fraiwan lung-sound dataset	T2	open Mendeley	lung-sound classification; disease-category classification	chest wall lung sounds	patient files P1-P112; exact complete clip count should be inspected after download	sound type and diagnosis; chest location metadata	sr c
8	RespiratoryDatabase@TR	T2	open Mendeley	multi-channel lung sounds; COPD severity; domain adaptation	multi-channel lung and heart/ lung recordings	75 subjects reported in associated article; 12-channel lung recordings reported	subject/file-level disease/severity; expert validation reported	sr c
9	HF_Lung_V2 / HF_Lung_V1_IP	T2	partially open; expanded labels partly controlled/ requested	expanded lung event detection	lung sounds	V2 paper reports train/test 10,554/3,403 files	onset-offset respiratory/ adventitious labels	sr c
10	HF_Tracheal_V1	T2	audio open; labels by request/form in some releases	breath phase and CAS detection	tracheal sounds	about 10k 15-s tracheal recordings in paper/repo descriptions; exact count should be verified after download	onset-offset I/E/ CAS labels	sr c

#	数据集/资源	层级	访问状态	主要任务路线	音频模态	关键规模	标签粒度	源
1 1	Coswara	T2	open GitHub	COVID-related cough/breath/speech analysis; symptom modeling	cough, breathing, speech, counting, vowels	Scientific Data paper reports 2,635 individuals; nine sound categories per participant	self-report/test-status subject labels; modality-level files	sr c
1 2	Cambridge COVID-19 Sounds	T2	DTA / academic access	COVID/symptom prediction; cough/breath/voice	cough, breathing, voice	53,449 samples; 552 h; 36,116 participants; 2,106 positive reported	self-reported testing status; participant-level metadata	sr c
1 3	DiCOVA Challenge datasets	T2	challenge-only / restricted challenge data	COVID detection from cough/breath/speech	cough, breathing, vowel, counting	Track 1: 1,040 subjects; Track 2: 990 subjects reported	binary subject-level COVID labels; challenge folds	sr c
1 4	ComParE 2021 COVID-19 Cough/Speech	T2	challenge subset	COVID detection from cough/speech	cough and speech	cough split 286/231/208; speech split 315/295/283 train/dev/test	binary subject-level COVID labels in challenge tracks	sr c
1 5	Sound-Dr	T2	open GitHub / paper-linked	COVID/symptom/anomaly prediction	cough, mouth breathing, nose breathing	3,930 recordings; 1,310 subjects	self-report COVID/symptom labels; modality-level files	sr c
1 6	smarty4covid	T2	public release with privacy exclusions	cough/breath/voice COVID-related modeling	cough, regular breath, deep breath, voice	18,265 recordings; 4,673 users	self-report health/COVID labels; expert-labeled subset	sr c
1 7	Virufy COVID-19 Open Cough Dataset	T2	open GitHub	PCR-linked cough; COVID cough analysis	cough	clinical subset described as 121 segmented cough samples from 16 patients	PCR-linked clinical labels; cough segments	sr c
1 8	Solicited TB Cough / CODA TB cough resources	T2	controlled/research access; exact public package varies	TB cough analysis; disease prediction	cough and possibly respiratory metadata	scale varies by release; exact public count not consistently verifiable from one official page	microbiology/clinical TB status where available	sr c
1 9	Audio-IMU Multimodal Cough Dataset	T2	open Dryad	cough event detection; wearable/bioacoustic multimodal sensing	wearable audio plus IMU	902 MB zip; 13 student volunteers; two chest-mounted microphones and MetaMotionS IMU	frame/clip labels: cough, speech, sneeze, deep breath, groan, laugh, speech_far, other	sr c
2 0	VocalSound	T2	open official/MIT/CSAIL + GitHub	cough/sneeze/sniff/throat-clearing pretraining; vocal-event detection	non-speech human vocalizations	21,024 recordings; 3,365 subjects	clip-level six-class vocalization labels; demographics/health-condition metadata	sr c
2 1	Pulmonary Sound Dataset / ALSD-Net data	T2	open Mendeley	normal/abnormal lung sound classification	electronic stethoscope lung sounds	Mendeley page reports normal 11.28%, abnormal 88.72%; full count requires download inspection	binary normal/abnormal; no event onset-offset found	sr c
2 2	Clinic Waiting Room Cough Dataset	T2	project/paper-linked; raw access unclear	clinic cough event detection	far-field clinic audio	3,930 manually annotated 3-s recordings from 54 waiting-room sessions; 348 h deployment validation reported	cough/non-cough clip labels; session/location metadata	sr c

#	数据集/资源	层级	访问状态	主要任务路线	音频模态	关键规模	标签粒度	源
2 3	DeepBreath / Pneumoscope pediatric auscultation dataset	T2	raw audio by reasonable request; code public	pediatric pathology/ pneumonia/ wheeze/ bronchiolitis modeling	pediatric lung auscultation	4,552 lung auscultation recordings; 572 pediatric outpatients; 35.9 h; six outpatient departments across five countries	diagnosis-level labels and attention/ localization outputs; by-site validation	sr c
2 4	Formosa Archive of Breath Sound (FABS)	T2	paper- described; public access not confirmed	ED lung-sound abnormality classification; label-noise analysis	lung sounds in emergency department context	5,238 annotated recordings; 1,985 patients; 14.6 h; 715 coarse crackles, 234 wheezes, 4,289 normal	physician- annotated normal/coarse crackle/wheeze labels	sr c
2 5	PKU Open Auscultation / Open Lung Sound resources	T2	open page/ GitHub; details need inspection	lung-sound classification; external validation	lung auscultation audio	exact public count/version unknown from high-level source	unknown / likely file-level lung sound labels	sr c
2 6	Resp-229K	T2	GitHub + Hugging Face announced; large download	dataset fusion; multimodal respiratory modeling; synthetic/LLM- narrative route	respiratory recordings with LLM-distilled clinical narratives	paper/repo report 229k recordings, 407+ h, about 70 GB; curated from ICBHI, SPRSound, UK COVID, COUGHVID, KAUH and HF Lung V1	source-derived labels plus LLM- distilled clinical narratives; semantics require audit	sr c
2 7	HLS-CMDS Simulated Heart/Lung Sound Dataset	T3	open Mendeley	pretraining; denoising; source separation; simulation sanity checks	clinical manikin heart/lung sounds	535 recordings; 50 heart, 50 lung, 145 mixed plus source signals; Littmann CORE, WAV/CSV	source/ separation labels; simulated context	sr c
2 8	UCL Speech Breath Monitoring / ComParE 2020 Breathing Sub-Challenge	T3	challenge / research dataset	speech-breath monitoring; breathing-from- speech auxiliary route	speech + respiratory belt signal	49 speakers; about 4 min per speaker; speech downsampled to 16 kHz; belt 25 Hz	continuous respiratory-belt target over speech	sr c
2 9	AudioSet respiratory- related ontology labels	T3	open metadata; YouTube availability varies	generic pretraining; weak auxiliary labels	web audio clips from YouTube	AudioSet full dataset is large; respiratory-related labels include cough, sneeze, sniff, breathing, snoring, wheeze-like labels depending ontology branch	weak clip-level labels, not clinical	sr c
3 0	AudioSet Cough class	T3	open metadata; YouTube availability varies	cough detector pretraining; weak positives	YouTube cough audio/video clips	ontology page reports 871 cough clips and about 2.4 h; 60 eval, 60 balanced train, 751 unbalanced train	weak clip-level Cough label	sr c
3 1	FSD50K / respiratory- sound slices	T3	open under source clip licenses	generic pretraining; weak respiratory event auxiliary	Freesound clips	51,197 clips; 200 classes; >100 h; respiratory-related subsets exist by labels/tags	weak labels from AudioSet ontology/ Freesound tags	sr c
3 2	ESC-50	T3	open GitHub; CC BY-NC for ESC-50	small generic cough/breath/ sneeze/snoring auxiliary	environmental sound clips	2,000 clips; 50 classes; 5 s; 40 examples/class; 44.1 kHz mono WAV	clip-level class labels; includes coughing, breathing, sneezing, snoring	sr c

#	数据集/资源	层级	访问状态	主要任务路线	音频模态	关键规模	标签粒度	源
3 3	cough-speech-sneeze / audEERING corpus	T3	available via conversion/ audb tooling	cough/sneeze/ speech auxiliary event modeling	YouTube-derived speech, cough, sneeze, silence	exact count should be inspected from audformat/ audb release	clip-level cough/ sneeze/speech/ silence labels	sr c
3 4	OpenSLR Nonverbal Vocalization Dataset	T3	open OpenSLR	nonverbal vocal event pretraining; cough/sneeze/ panting/yawn auxiliaries	crowdsourced nonverbal vocalizations	scale not transcribed here; official page lists 16 nonverbal classes	clip-level labels include coughing, yawning, throat clearing, sighing, panting, sneezing etc.	sr c
3 5	FluSense hospital audio dataset	T3	paper/project; public raw access unclear	population-level cough/symptom trend monitoring	waiting-room audio/video/ sensor streams	paper reports hospital waiting-area deployment; exact raw release not found	cough/sneeze/ audio-event and aggregate illness trend labels	sr c
3 6	NoCoCoDa	T3	dataset page / paper-linked	non-COVID cough controls; cough negative/ positive balance	cough audio	exact scale/access terms need inspection from dataset page	cough and non-COVID status labels	sr c
3 7	VOICED Database	T3	open PhysioNet	voice pathology; speech-health background	sustained vowel voice samples	208 clinically verified voice samples; 150 pathological, 58 healthy	speaker-level pathology labels; demographics and VHI/RSI questionnaire metadata	sr c
3 8	Saarbruecken Voice Database (SVD)	T3	open web interface/ download terms vary	voice pathology; speech-health auxiliary	sustained vowels and speech tasks	large voice pathology database; exact current count should be checked from official site	speaker/ pathology labels; speech task metadata	sr c
3 9	Bridge2AI-Voice Adult Dataset	T3	PhysioNet derived features open/ controlled; raw audio via Synapse/DACO	speech-health and multimodal health monitoring; respiratory/ voice-disorder context	voice-derived features; raw audio controlled	PhysioNet v3.1.0 published May 2026; raw audio not in PhysioNet release	derived features, questionnaires, clinical/ demographic metadata	sr c
4 0	Bridge2AI-Voice Pediatric Dataset	T3	PhysioNet derived features; raw access controlled	pediatric speech-health monitoring; respiratory/voice biomarker context	voice-derived features/ spectrograms and clinical metadata	PhysioNet v1.1.0 published May 2026	derived features, spectrograms, questionnaire and clinical metadata	sr c
4 1	OPERA benchmark/ resource	T3	open code; underlying datasets have mixed access/ licenses	respiratory acoustic foundation-model benchmarking	fused respiratory acoustic tasks	GitHub/OpenReview report about 136k samples, 440 h, 19 tasks	task labels inherited from constituent datasets	sr c
4 2	Google HeAR / Health Acoustic Representations	T3	model gated on Hugging Face; paper/ model docs public	frozen embedding baseline; health-audio representation	generic health acoustic embeddings	paper/model docs report 313M two-second clips; evaluation on 33 tasks across six datasets	not a dataset label source; representation model	sr c

#	数据集/资源	层级	访问状态	主要任务路线	音频模态	关键规模	标签粒度	源
4 3	RALE Lung Sounds Repository	T4	web-listened education/ repository; research download/ metadata limited	reference examples only	respiratory sounds	official page groups recordings into Normal, Wheezes, Crackles and Other; exact research-ready metadata unknown	coarse category labels	sr c
4 4	SNUCH Pediatric Lung Sound Dataset	T4	private / not public	pediatric lung-sound method evidence	pediatric lung sounds	paper reports 680 eligible training/internal-validation clips and 90 prospective-validation clips	normal/crackle/ wheeze labels by pediatric pulmonologists	sr c
4 5	MIT Open Voice / cough AI preliminary resources	T4	project/ repository status unclear	speech/cough health monitoring background	voice/cough depending release	unknown / not found as stable raw-audio dataset	unknown	sr c
4 6	MEEI Voice Disorders Database	T4	commercial/ restricted legacy dataset	voice pathology background	voice/speech	secondary papers report >1,400 recordings or 657 pathological + 53 normal subsets; exact accessible package varies	voice pathology labels	sr c
4 7	PEEP / respiratory-support physiological datasets without raw audio	T4	open/non-audio; not central	clinical context only	not raw respiratory audio	varies	not acoustic labels	sr c
4 8	Open-source AI Stethoscope / COPD HeAR demo resources	T4	GitHub/demo; paper/dataset status not fully verified	method lead only	stethoscope lung sounds	repo claims COPD AUROC but source dataset/access need verification	unknown	sr c
4 9	Kaggle ICBHI respiratory-sound mirrors	T4	open mirrors; often unofficial	convenient download only, not primary source	mirrored ICBHI lung sounds	920 recordings when faithful; mirror metadata may drift	inherits ICBHI labels	sr c
5 0	YouTube/ Freesound cough compilation datasets	T4	varies; often weakly documented	negative/ auxiliary only	web cough/ sneeze/breath clips	unknown; many mirrors duplicate AudioSet/FSD/ESC	weak clip tags	sr c
5 1	Simulated/ body-sound teaching libraries	T4	varies; educational license	qualitative listening/ reference only	lung/heart/ respiratory examples	unknown / not research-ready unless metadata available	coarse teaching labels	sr c
5 2	General speech corpora sometimes used for health-audio pretraining	T4	open but weak connection	pretraining only; not respiratory-health evidence	ordinary speech	large but not respiratory-specific	transcript/ speaker labels, not disease labels	sr c

4. Relevance Tier Summary

4.1 Tier 1: 强直接适配

Tier 1 数据集是短期实验最应优先投入的资源：**ICBHI 2017**、**HF_Lung_V1**、**SPRSound**、**COUGHVID**、**UK COVID-19 Vocal Audio**、**Corp Dataset**。其中 ICBHI/HF/SPRSound 覆盖肺音与呼吸事件，COUGHVID/Corp 覆盖咳嗽事件与咳嗽质量，UK Vocal 提供更可靠的 PCR-linked disease label。Tier 1 内部也不能无条件合并：ICBHI 的 cycle-level crackle/wheeze、HF 的 onset-offset CAS/DAS、SPRSound 的 pediatric event ontology、COUGHVID 的 self-report metadata、UK Vocal 的 PCR-linked labels 属于不同语义层。

4.2 Tier 2: 有用但有 caveats

Tier 2 包括 KAUH、RespiratoryDatabase@TR、HF_Lung_V2、HF_Tracheal、Coswara、Cambridge、DiCOVA、ComParE COVID、Sound-Dr、smarty4covid、Virufy、TB cough resources、Audio-IMU Cough、VocalSound、Pulmonary Sound、Clinic Waiting Room Cough、DeepBreath、FABS、Resp-229K 等。它们适合外部验证、扩充模态、补充咳嗽/呼吸/语音健康背景或方法复现，但每个都有明确限制：访问受限、自报告标签、私有数据、儿童/成人 mismatch、非临床志愿者、或 derived/LLM labels。

4.3 Tier 3: 辅助/背景证据

Tier 3 是表示学习和方法学背景：AudioSet、FSD50K、ESC-50、OpenSLR Nonverbal、cough-speech-sneeze、UCL-SBM/ComParE Breathing、VOICED、SVD、Bridge2AI-Voice、OPERA、HeAR 等。它们适合用作 frozen embedding、pretraining、negative set、domain-adaptation background 或 speech-health route；不应用来直接声明 respiratory disease detection 能力。

4.4 Tier 4: 仅提及/低可行动性

Tier 4 包括 RALE、SNUCH private pediatric dataset、MEEI、MIT Open Voice、Kaggle mirrors、teaching libraries 等。它们可以提供背景或联系线索，但不应作为短期可复现实验主数据。

5. Task-Route Map

5.1 Event detection

优先数据集：ICBHI、HF_Lung_V1、SPRSound、Corp Dataset、Audio-IMU Cough、Clinic Waiting Room Cough。任务应拆为：lung cycle classification、adventitious event onset-offset、breath phase segmentation、cough event detection、cough boundary localization。短期最合理的论文目标是 **event-level robust modeling**，而不是直接 clinical diagnosis。

5.2 Disease prediction

优先数据集：UK COVID Vocal、DiCOVA、ComParE COVID、Cambridge、Coswara、Sound-Dr、Virufy、TB cough resources、DeepBreath/FABS self-collected papers。需要严格区分 PCR-linked、self-report、challenge-derived、diagnosis-level 和 symptom-level labels。疾病预测路线必须报告 symptom-only 或 metadata-only baseline，尤其在 COVID 音频上。

5.3 Cough/breath analysis

优先数据集：COUGHVID、UK Vocal cough/exhalation、Coswara、Audio-IMU Cough、VocalSound、cough-speech-sneeze、OpenSLR NVD、AudioSet Cough、Clinic Waiting Room Cough、Corp Dataset。推荐先做 cough/non-cough、quality filtering、segmentation、artifact rejection，再进入 disease-label modeling。

5.4 Speech-health assessment

优先资源：Bridge2AI-Voice Adult/Pediatric、VOICED、SVD、Cambridge/Coswara/UK Vocal speech components（注意 UK open edition不包含 speech）、MEEI 作为受限背景。语音健康路线应和 lung/cough 路线分开写，关注 voice biomarker、vocal fold pathology、respiratory/speech-breath coupling 和 telehealth monitoring，不把 voice pathology label 当作 lung disease label。

5.5 Dataset fusion / domain adaptation

优先资源：OPERA、Resp-229K、HeAR、AudioSet/FSD50K/ESC-50、ICBHI/HF/SPRSound/KAUH cross-dataset。Fusion 的核心风险是 label semantics 与 source leakage。若 Resp-229K 使用 ICBHI/HF/SPRSound/UK/COUGHVID/KAUH 源数据，则下游评测必须排除源重叠或至少报告 contamination audit。

5.6 Weak auxiliary evidence

AudioSet、FSD50K、ESC-50、OpenSLR、VocalSound、cough-speech-sneeze 可以提供弱标签和预训练信号，但其 web-audio 标签是非临床标签。它们适合降低 false positives、学习 human non-speech events 或作为 frozen embedding baseline 的背景，不应作为临床结论证据。

6. Dataset Deep-Dive Profiles

6.1 ICBHI 2017 Respiratory Sound Database

- **Tier:** Tier 1.
- **Official URL / citation:** ICBHI challenge/mirror and the database paper: <https://ai4eu.dei.uc.pt/respiratory-sounds-dataset/>, https://bhichallenge.med.auth.gr/ICBHI_2017_Challenge, and Rocha et al., 2019.
- **Access / license:** open through official/mirror pages; some mirrors state CC0/public-domain style reuse, but the exact license should be preserved from the accessed package.
- **Task route:** core event detection and lung-sound classification.
- **Audio modality/context:** heterogeneous chest respiratory recordings collected with multiple devices and chest positions.
- **Technical format:** WAV files in the common release; sampling rate/device varies by original acquisition; exact per-file specs should be read from WAV headers.
- **Scale:** 920 recordings, 6,898 respiratory cycles, 126 subjects, about 5.5 hours.

- **Age/demographics:** adult and pediatric ages appear in metadata; exact age distribution should be computed from metadata.
- **Labels:** respiratory-cycle annotations for normal, crackle, wheeze, and both crackle+wheeze; diagnosis metadata include COPD/asthma/bronchiectasis/pneumonia/URTI/LRTI/healthy in common metadata.
- **Granularity:** cycle-level acoustic-event labels; subject/recording-level metadata; expert/clinician annotation in dataset paper.
- **Splits/protocol:** official challenge split exists. Many papers use custom or random cycle splits; those are not directly comparable.
- **Leakage risks:** cycle-level random split can place cycles from the same patient or recording in train/test. Use patient-independent split.
- **Recommended use:** first public lung-sound benchmark; use official split and ICBHI score, sensitivity, specificity, macro-F1; include external validation on HF_Lung or SPRSound if claiming robustness.

6.2 HF_Lung_V1

- **Tier:** Tier 1.
- **Official URL / citation:** https://gitlab.com/techsupportHF/HF_Lung_V1; Hsu et al., PLOS ONE 2021.
- **Access / license:** open GitLab release; paper/repo state CC BY-NC 4.0 in many derived resources.
- **Task route:** event detection, breath-phase segmentation, continuous adventitious sound (CAS) and discontinuous adventitious sound (DAS) detection.
- **Audio modality/context:** lung sound recordings from electronic stethoscope settings; 15-second clips.
- **Technical format:** paper reports 4 kHz sampling, 16-bit WAV, mono.

- **Scale:** 9,765 15-second recordings; 34,095 inhalation labels; 18,349 exhalation labels; 13,883 CAS labels; 15,606 DAS labels.
- **Subjects/metadata:** exact subject count and demographics should be read from metadata; disease labels are not the primary strength.
- **Labels:** onset-offset annotations for inhalation, exhalation, CAS, DAS.
- **Granularity:** event-level onset-offset; expert annotation.
- **Splits/protocol:** user-defined; subject independence must be enforced if subject IDs are available.
- **Leakage risks:** clip-level split across repeated recordings can inflate event metrics.
- **Recommended use:** strongest open event-level lung-sound corpus for model development; useful for boundary detection and temporal models before disease prediction.

6.3 SPRSound

- **Tier:** Tier 1.
- **Official URL / citation:** <https://github.com/SJTU-YONGFU-RESEARCH-GRP/SPRSound>; IEEE TBioCAS 2022 database paper and challenge pages.
- **Access / license:** open GitHub/challenge releases; license terms should be checked in the release.
- **Task route:** pediatric respiratory sound event detection and classification.
- **Audio modality/context:** pediatric lung sounds, collected from children aged roughly 1 month to 18 years using Yunting model II stethoscope according to the database paper/challenge material.
- **Technical format:** WAV files; exact sample rate/bit depth should be verified from release.

- **Scale:** 2022 challenge release lists 1,949 training and 734 test records; original paper reports 2,683 records, 292 participants, and 9,089 events.
- **Labels:** normal, poor-quality, CAS types and DAS types in challenge/task definitions; includes wheeze, stridor, rhonchi, coarse/fine crackle in event labels.
- **Granularity:** record-level labels and event onset-offset annotations.
- **Splits/protocol:** challenge train/test splits; later evaluation sets exist.
- **Leakage risks:** pediatric subject independence and site independence need explicit checking.
- **Recommended use:** essential pediatric external domain; do not mix with adult stethoscope data without age-domain analysis.

6.4 COUGHVID

- **Tier:** Tier 1.
- **Official URL / citation:** EPFL dataset page, Zenodo release, Orlandic et al., Scientific Data 2021.
- **Access / license:** open Zenodo; license is generally CC BY 4.0 in official releases, but downstream derivatives should preserve metadata and consent constraints.
- **Task route:** cough quality control, cough detection/segmentation, weak COVID/symptom analysis, representation learning.
- **Audio modality/context:** crowdsourced cough through web/mobile; uncontrolled devices and environments.
- **Technical format:** web audio; common release uses Opus/webm/JSON metadata; exact specs vary by uploaded file.

- **Scale:** over 30k cough recordings in later releases; Scientific Data paper describes 25k+ crowdsourced recordings and 2,800+ expert-labeled coughs.
- **Labels:** expert labels for cough/no cough/quality/diagnostic attributes; self-reported COVID and symptoms.
- **Granularity:** file-level labels, metadata-level symptoms, some segmentation from derived tools.
- **Splits/protocol:** no single universal benchmark; papers use incompatible filtering and splits.
- **Leakage risks:** repeated users, upload date, country/device and self-report confounding.
- **Recommended use:** strong cough pretraining and cough-quality route; avoid treating self-reported COVID as equivalent to PCR-linked labels.

6.5 UK COVID-19 Vocal Audio / COVID-19 Vocal Audio

- **Tier:** Tier 1.
- **Official URL / citation:** Zenodo release and Alan Turing Institute GitHub; Scientific Data 2024 data paper.
- **Access / license:** open cough/exhalation release under UK Open Government Licence v3.0; speech was collected but excluded from open release because it is more identifiable.
- **Task route:** PCR-linked COVID disease prediction; cough and exhalation analysis; negative-result benchmarking against symptom models.
- **Audio modality/context:** national-scale app/web data with cough, exhalation and speech collection; open edition contains cough/exhalation.

- **Technical format:** WAV files plus CSV metadata; 53.7 GB open release reported.
- **Scale:** 72,999 participants and 70,794 PCR-linked participants reported in official release; exact usable count depends on quality filters.
- **Labels:** PCR-linked SARS-CoV-2 status; symptoms and demographics.
- **Granularity:** subject/session-level test labels; file-level modality.
- **Splits/protocol:** official baseline code exists; strong studies compare audio-only/audio+symptom/symptom-only.
- **Leakage risks:** recruitment, test-date, device, symptom and demographic confounding; leakage if multiple submissions by same participant are mixed.
- **Recommended use:** best public disease-label dataset among COVID audio resources; use it to test whether audio adds value beyond symptoms, not just to maximize audio-only AUC.

6.6 Corp Dataset

- **Tier:** Tier 1.
- **Official URL / citation:** C-BiLSTM cough event boundary paper and Tongji MARI dataset/project page.
- **Access / license:** access path unclear from public page; verify by contacting maintainers before planning reproducible benchmark experiments.
- **Task route:** long-duration cough event detection and boundary localization.
- **Audio modality/context:** respiratory-disease patient recordings; long-duration cough monitoring rather than short solicited cough.

- **Technical format:** paper reports 44.1 kHz / 192 kbps; exact file packaging unknown.
- **Scale:** 168 h audio, 9,969 cough events, 42 patients.
- **Labels:** cough event boundaries; disease known at subject level.
- **Granularity:** onset-offset cough events; subject-level disease.
- **Splits/protocol:** paper-defined experiments; patient-independent setup should be required for new work.
- **Leakage risks:** segment-level random split across one long recording or patient can inflate results.
- **Recommended use:** excellent target for event detection if access is obtained; otherwise include as a method-comparison reference, not as a guaranteed dataset.

6.7 KAUH / Fraiwan lung-sound dataset

- **Tier:** Tier 2.
- **Official URL / citation:** Mendeley Data [jwyy9np4gv/3](https://data.mendeley.com/datasets/jwyy9np4gv/3).
- **Access / license:** open Mendeley; license must be read from dataset page/package.
- **Task route:** lung-sound event/file classification and diagnosis-conditioned external validation.
- **Audio modality/context:** electronic stethoscope recordings from chest wall positions.
- **Technical format:** WAV; exact sample rate/bit depth should be extracted from files.
- **Scale:** filename scheme P1-P112 suggests 112 patient IDs; exact file count should be computed after download.

- **Labels:** annotation includes sound type I/E/W/C/N, diagnosis such as asthma/COPD/bronchiectasis/heart failure/lung fibrosis and chest location.
- **Granularity:** file-level and metadata labels; onset-offset not verified.
- **Caveat:** diagnosis labels and event labels are not equivalent; patient-level split is mandatory.
- **Recommended use:** external validation and small-domain transfer for ICBHI/HF models.

6.8 RespiratoryDatabase@TR

- **Tier:** Tier 2.
- **Official URL / citation:** Mendeley p9z4h98s6j/1 and associated DergiPark article.
- **Access / license:** open Mendeley; license terms should be checked before redistribution.
- **Task route:** multi-channel lung sound, COPD severity, domain adaptation and source-location experiments.
- **Audio modality/context:** multi-channel lung sound recordings and heart/lung recordings.
- **Scale:** associated article reports 75 subjects; detailed file/channel counts require download inspection.
- **Labels:** subject-level condition/severity and recording metadata; exact event annotations unknown.
- **Granularity:** likely file/subject-level rather than onset-offset.
- **Caveat:** task definition differs from ICBHI/HF event detection.

- **Recommended use:** external validation for disease/severity and multi-channel representation, not primary event benchmark.

6.9 HF_Lung_V2 / HF_Lung_V1_IP

- **Tier:** Tier 2.
- **Source:** Applied Sciences 2022 article and GitLab V1_IP release.
- **Access:** partial/open-plus-request depending version; not all labels may be unrestricted.
- **Route:** event-level respiratory/adventitious detection and robustness benchmarking.
- **Scale/labels:** paper reports 10,554 train and 3,403 test files in V2; event labels are similar in spirit to HF_Lung_V1 but version-specific.
- **Recommended use:** useful if access is granted; do not mix V1/V2 labels without documenting taxonomy differences.

6.10 HF_Tracheal_V1

- **Tier:** Tier 2.
- **Source:** arXiv and GitLab.
- **Access:** audio is public in repo; labels may require form/request depending release.
- **Route:** tracheal breath-phase detection and CAS detection.
- **Scale:** about 10k 15-second tracheal recordings are described; exact count should be verified after download.
- **Caveat:** tracheal signals are not equivalent to chest-wall lung sounds; transfer function and event semantics differ.

- **Recommended use:** breath-phase modeling, domain shift from lung to tracheal audio, auxiliary temporal segmentation.

6.11 Coswara

- **Tier:** Tier 2.
- **Source:** IISc LEAP GitHub and Scientific Data/arXiv paper.
- **Access:** open GitHub.
- **Route:** multimodal cough/breath/speech disease prediction and representation learning.
- **Audio:** multiple modalities per participant: cough, breathing, speech/counting/vowels.
- **Scale:** paper reports 2,635 individuals and nine sound categories.
- **Labels:** self-reported/test-status metadata; not as strong as PCR-linked registry labels.
- **Caveat:** severe label and recruitment confounds; repeated updates can change counts.
- **Recommended use:** multimodal pipeline proof-of-concept and weakly supervised pretraining; not final clinical evidence.

6.12 Cambridge COVID-19 Sounds

- **Tier:** Tier 2.
- **Source:** OpenReview dataset paper, DTA information page, and npj Digital Medicine analysis.
- **Access:** DTA/controlled academic access.
- **Route:** COVID/symptom prediction from cough, breathing and voice.

- **Scale:** paper reports 53,449 samples, 552 h, 36,116 participants, 2,106 positives.
- **Labels:** self-reported COVID/test status and symptoms.
- **Caveat:** data-access friction and self-report confounding; the npj Digital Medicine paper is important because it highlights bias and realistic performance issues.
- **Recommended use:** reproducible only after DTA; good for bias-analysis route.

6.13 DiCOVA Challenge datasets

- **Tier:** Tier 2.
- **Source:** DiCOVA challenge website and challenge papers.
- **Access:** challenge-only/restricted.
- **Route:** COVID detection from cough/breath/speech.
- **Scale:** Track 1 cough subset is commonly reported as 1,040 subjects; Track 2 as 990 subjects from Coswara-derived data.
- **Labels:** binary subject-level COVID labels with official folds.
- **Caveat:** comparable only within a specific DiCOVA track/fold/metric.
- **Recommended use:** benchmark reproduction if access is obtained; otherwise cite as historical challenge.

6.14 ComParE 2021 COVID-19 Cough/Speech

- **Tier:** Tier 2.
- **Source:** Interspeech/ComParE challenge summary.
- **Access:** challenge data; public availability depends on challenge terms.
- **Route:** COVID detection from cough and speech.

- **Scale:** challenge table reports cough train/dev/test 286/231/208 and speech train/dev/test 315/295/283.
- **Labels:** binary COVID label; subject-level.
- **Caveat:** small and challenge-specific; cannot be merged directly with other COVID audio datasets.

6.15 Sound-Dr

- **Tier:** Tier 2.
- **Source:** GitHub and PHM Asia paper.
- **Access:** open repo.
- **Route:** COVID/anomaly prediction from cough, mouth-breathing and nose-breathing.
- **Scale:** 3,930 recordings from 1,310 subjects.
- **Labels:** self-report COVID/symptom labels and modality labels.
- **Caveat:** self-report and small/moderate scale; exact splits should be reproduced from paper/repo.

6.16 smarty4covid

- **Tier:** Tier 2.
- **Source:** Scientific Data 2023.
- **Access:** public release with privacy exclusions.
- **Route:** COVID-related cough/breath/voice modeling and audio knowledge-base work.
- **Scale:** paper reports 18,265 recordings from 4,673 users, including cough, regular breathing, deep breathing and voice.

- **Labels:** self-reported health/COVID metadata; an expert-labeled subset exists.
- **Caveat:** voice privacy exclusions and self-report labels.

6.17 Virufy COVID-19 Open Cough Dataset

- **Tier:** Tier 2.
- **Source:** Virufy GitHub.
- **Access:** open GitHub.
- **Route:** PCR-linked cough classification and small clinical cough analysis.
- **Scale:** clinical subset described in repository/materials as 121 segmented cough samples from 16 patients; exact version should be pinned.
- **Labels:** PCR-linked positives/negatives where available.
- **Caveat:** small sample size; version drift; not sufficient alone for strong claims.

6.18 Solicited TB cough / CODA TB cough resources

- **Tier:** Tier 2.
- **Source:** cough-disease literature and controlled-access challenge/dataset references; no single stable open primary package was verified in this pass.
- **Access:** controlled or unclear.
- **Route:** TB cough disease prediction.
- **Labels:** strongest variants use microbiology/clinical TB labels; exact granularity depends on release.

- **Caveat:** do not use unless access terms and label provenance are confirmed.
- **Recommended use:** long-term disease route after access; not a short-term public benchmark.

6.19 Audio-IMU Multimodal Cough Dataset

- **Tier:** Tier 2.
- **Source:** Dryad DOI [10.5061/dryad.0p2ngf26t](https://doi.org/10.5061/dryad.0p2ngf26t).
- **Access:** open Dryad, about 902 MB.
- **Route:** cough event detection, wearable multimodal fusion, out-of-distribution robustness.
- **Audio/context:** two chest-mounted microphones derived from Tozo T10 Bluetooth earbuds plus MetaMotionS r1 9-axis IMU.
- **Scale:** 13 participants, controlled volunteer recordings.
- **Labels:** cough, speech, sneeze, deep breath, groan, laugh, far speech and other.
- **Caveat:** healthy/student volunteer context; not disease prediction.
- **Recommended use:** excellent auxiliary cough/non-cough event detection and multimodal baseline.

6.20 VocalSound

- **Tier:** Tier 2.
- **Source:** MIT/CSAIL official page, GitHub and paper.
- **Access:** open for research under dataset terms; verify license before redistribution.

- **Route:** cough/sneeze/sniff/throat-clearing pretraining and vocal event detection.
- **Scale:** official page reports 21,024 crowdsourced recordings from 3,365 subjects.
- **Labels:** laughter, sigh, cough, throat clearing, sneeze, sniff; age, gender, native language, country and health-condition metadata.
- **Caveat:** not a respiratory-disease dataset; use as auxiliary representation/negative/foreground event corpus.

6.21 Pulmonary Sound Dataset / ALSD-Net data

- **Tier:** Tier 2.
- **Source:** Mendeley dataset and ALSD-Net citation.
- **Access:** open Mendeley.
- **Route:** normal/abnormal lung sound classification.
- **Labels:** binary normal/abnormal; Mendeley page reports strong class imbalance.
- **Caveat:** no verified onset-offset event labels; exact data count and device specs require package inspection.
- **Recommended use:** binary abnormality external validation, with caution around imbalance.

6.22 Clinic Waiting Room Cough Dataset

- **Tier:** Tier 2.
- **Source:** Scientific Reports cough detection paper/project.
- **Access:** raw data not confirmed public; model/paper definition is clear enough to include in method map.

- **Route:** cough event detection in real clinical acoustic environments.
- **Scale:** paper reports 3,930 manually annotated 3-second recordings from 54 sessions and validation on 348 hours of waiting-room audio.
- **Labels:** cough/non-cough clips and deployment event labels.
- **Caveat:** raw data access and session-level split reproducibility need confirmation.

6.23 DeepBreath / Pneumoscope pediatric auscultation dataset

- **Tier:** Tier 2.
- **Source:** npj Digital Medicine 2023 paper and public GitHub.
- **Access:** code is public; raw audio available only on reasonable request because consent/privacy do not allow open release.
- **Route:** self-collected but reproducible-method pediatric auscultation paper; pneumonia/wheezing/bronchiolitis modeling.
- **Scale:** GitHub/paper report 4,552 lung auscultation recordings from 572 pediatric outpatients, 35.9 hours, six outpatient departments in five countries.
- **Labels:** diagnosis/pathology labels; train on two sites with internal and external validation in the released code workflow.
- **Caveat:** not immediately downloadable; still useful as a reproducible architecture/protocol reference.

6.24 Formosa Archive of Breath Sound (FABS)

- **Tier:** Tier 2.
- **Source:** JMIR AI 2025 paper.
- **Access:** public raw-data access not confirmed.

- **Route:** direct lung-sound abnormality classification with multi-annotator physician labels; useful self-collected paper if model definition is followed.
- **Scale:** paper reports 5,238 annotated recordings, 1,985 patients and 14.6 h, with 715 coarse crackles, 234 wheezes and 4,289 normal recordings.
- **Labels:** seven senior physicians annotated/validated labels according to paper.
- **Caveat:** ED-specific domain, access unclear and class imbalance.

6.25 PKU Open Auscultation / open lung-sound resources

- **Tier:** Tier 2 if official package is verified; otherwise Tier 4.
- **Source:** repository leads exist, but official stable citation and metadata still require verification.
- **Route:** potential external lung-sound validation.
- **Caveat:** do not include in experiments until exact source, license, subject count, labels and split rules are confirmed.

6.26 Resp-229K

- **Tier:** Tier 2.
- **Source:** Resp-Agent arXiv/OpenReview/GitHub.
- **Access:** GitHub and Hugging Face are announced; repo reports about 70 GB.
- **Route:** dataset fusion/domain adaptation, respiratory foundation-model pretraining, multimodal clinical-narrative modeling.
- **Scale:** paper/repo report 229k audio files and 407+ hours curated from ICBHI, SPRSound, UK COVID, COUGHVID, KAUH and HF Lung V1.

- **Labels:** source-derived labels plus LLM-distilled clinical narratives.
- **Caveat:** label semantics are derived and partly generated; must audit source leakage if downstream evaluation uses constituent datasets.

6.27 HLS-CMDS Simulated Heart/Lung Sound Dataset

- **Tier:** Tier 3.
- **Source:** <https://data.mendeley.com/datasets/pxg5f7j5kd/1>.
- **Access:** open Mendeley.
- **Route:** pretraining; denoising; source separation; simulation sanity checks.
- **Audio modality:** clinical manikin heart/lung sounds.
- **Scale / format:** 535 recordings; 50 heart, 50 lung, 145 mixed plus source signals; Littmann CORE, WAV/CSV.
- **Labels / granularity:** source/separation labels; simulated context.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.28 UCL Speech Breath Monitoring / ComParE 2020 Breathing Sub-Challenge

- **Tier:** Tier 3.

- **Source:** https://www.isca-archive.org/interspeech_2020/schuller20_interspeech.html.
- **Access:** challenge / research dataset.
- **Route:** speech-breath monitoring; breathing-from-speech auxiliary route.
- **Audio modality:** speech + respiratory belt signal.
- **Scale / format:** 49 speakers; about 4 min per speaker; speech downsampled to 16 kHz; belt 25 Hz.
- **Labels / granularity:** continuous respiratory-belt target over speech.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.29 AudioSet respiratory-related ontology labels

- **Tier:** Tier 3.
- **Source:** <https://research.google.com/audioset/ontology/index.html>.
- **Access:** open metadata; YouTube availability varies.
- **Route:** generic pretraining; weak auxiliary labels.
- **Audio modality:** web audio clips from YouTube.
- **Scale / format:** AudioSet full dataset is large; respiratory-related labels include cough, sneeze, sniff, breathing, snoring, wheeze-like labels depending on ontology branch.

- **Labels / granularity:** weak clip-level labels, not clinical.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.30 AudioSet Cough class

- **Tier:** Tier 3.
- **Source:** https://research.google.com/audioset/ontology/cough_1.html.
- **Access:** open metadata; YouTube availability varies.
- **Route:** cough detector pretraining; weak positives.
- **Audio modality:** YouTube cough audio/video clips.
- **Scale / format:** ontology page reports 871 cough clips and about 2.4 h; 60 eval, 60 balanced train, 751 unbalanced train.
- **Labels / granularity:** weak clip-level Cough label.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.31 FSD50K / respiratory-sound slices

- **Tier:** Tier 3.
- **Source:** <https://zenodo.org/records/4060432>.
- **Access:** open under source clip licenses.
- **Route:** generic pretraining; weak respiratory event auxiliary.
- **Audio modality:** Freesound clips.
- **Scale / format:** 51,197 clips; 200 classes; >100 h; respiratory-related subsets exist by labels/tags.
- **Labels / granularity:** weak labels from AudioSet ontology/Freesound tags.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.32 ESC-50

- **Tier:** Tier 3.
- **Source:** <https://github.com/karolpiczak/ESC-50>.
- **Access:** open GitHub; CC BY-NC for ESC-50.
- **Route:** small generic cough/breath/sneeze/snoring auxiliary.
- **Audio modality:** environmental sound clips.
- **Scale / format:** 2,000 clips; 50 classes; 5 s; 40 examples/class; 44.1 kHz mono WAV.

- **Labels / granularity:** clip-level class labels; includes coughing, breathing, sneezing, snoring.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.33 cough-speech-sneeze / audEERING corpus

- **Tier:** Tier 3.
- **Source:** <https://audeering.github.io/datasets/datasets/cough-speech-sneeze.html>.
- **Access:** available via conversion/audb tooling.
- **Route:** cough/sneeze/speech auxiliary event modeling.
- **Audio modality:** YouTube-derived speech, cough, sneeze, silence.
- **Scale / format:** exact count should be inspected from audformat/audb release.
- **Labels / granularity:** clip-level cough/sneeze/speech/silence labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.34 OpenSLR Nonverbal Vocalization Dataset

- **Tier:** Tier 3.
- **Source:** <https://www.openslr.org/99/>.
- **Access:** open OpenSLR.
- **Route:** nonverbal vocal event pretraining; cough/sneeze/panting/yawn auxiliaries.
- **Audio modality:** crowdsourced nonverbal vocalizations.
- **Scale / format:** scale not transcribed here; official page lists 16 nonverbal classes.
- **Labels / granularity:** clip-level labels include coughing, yawning, throat clearing, sighing, panting, sneezing etc..
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.35 FluSense hospital audio dataset

- **Tier:** Tier 3.
- **Source:** <https://dl.acm.org/doi/10.1145/3381014>.
- **Access:** paper/project; public raw access unclear.
- **Route:** population-level cough/symptom trend monitoring.
- **Audio modality:** waiting-room audio/video/sensor streams.

- **Scale / format:** paper reports hospital waiting-area deployment; exact raw release not found.
- **Labels / granularity:** cough/sneeze/audio-event and aggregate illness trend labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.36 NoCoCoDa

- **Tier:** Tier 3.
- **Source:** <https://www.nature.com/articles/s41597-023-02646-6>.
- **Access:** dataset page / paper-linked.
- **Route:** non-COVID cough controls; cough negative/positive balance.
- **Audio modality:** cough audio.
- **Scale / format:** exact scale/access terms need inspection from dataset page.
- **Labels / granularity:** cough and non-COVID status labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.

- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.37 VOICED Database

- **Tier:** Tier 3.
- **Source:** <https://physionet.org/content/voiced/>.
- **Access:** open PhysioNet.
- **Route:** voice pathology; speech-health background.
- **Audio modality:** sustained vowel voice samples.
- **Scale / format:** 208 clinically verified voice samples; 150 pathological, 58 healthy.
- **Labels / granularity:** speaker-level pathology labels; demographics and VHI/RSI questionnaire metadata.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.38 Saarbruecken Voice Database (SVD)

- **Tier:** Tier 3.
- **Source:** <https://stimmdb.coli.uni-saarland.de/>.
- **Access:** open web interface/download terms vary.

- **Route:** voice pathology; speech-health auxiliary.
- **Audio modality:** sustained vowels and speech tasks.
- **Scale / format:** large voice pathology database; exact current count should be checked from official site.
- **Labels / granularity:** speaker/pathology labels; speech task metadata.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.39 Bridge2AI-Voice Adult Dataset

- **Tier:** Tier 3.
- **Source:** <https://physionet.org/content/b2ai-voice/>.
- **Access:** PhysioNet derived features open/controlled; raw audio via Synapse/DACO.
- **Route:** speech-health and multimodal health monitoring; respiratory/voice-disorder context.
- **Audio modality:** voice-derived features; raw audio controlled.
- **Scale / format:** PhysioNet v3.1.0 published May 2026; raw audio not in PhysioNet release.
- **Labels / granularity:** derived features, questionnaires, clinical/demographic metadata.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this

report treats the dataset as auxiliary or low-actionability rather than core evidence.

- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.40 Bridge2AI-Voice Pediatric Dataset

- **Tier:** Tier 3.
- **Source:** <https://physionet.org/content/b2ai-voice-pediatric/>.
- **Access:** PhysioNet derived features; raw access controlled.
- **Route:** pediatric speech-health monitoring; respiratory/voice biomarker context.
- **Audio modality:** voice-derived features/spectrograms and clinical metadata.
- **Scale / format:** PhysioNet v1.1.0 published May 2026.
- **Labels / granularity:** derived features, spectrograms, questionnaire and clinical metadata.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.41 OPERA benchmark/resource

- **Tier:** Tier 3.
- **Source:** <https://github.com/evelyn0414/OPERA>.
- **Access:** open code; underlying datasets have mixed access/licenses.
- **Route:** respiratory acoustic foundation-model benchmarking.
- **Audio modality:** fused respiratory acoustic tasks.
- **Scale / format:** GitHub/OpenReview report about 136k samples, 440 h, 19 tasks.
- **Labels / granularity:** task labels inherited from constituent datasets.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.42 Google HeAR / Health Acoustic Representations

- **Tier:** Tier 3.
- **Source:** <https://huggingface.co/google/hear>.
- **Access:** model gated on Hugging Face; paper/model docs public.
- **Route:** frozen embedding baseline; health-audio representation.
- **Audio modality:** generic health acoustic embeddings.
- **Scale / format:** paper/model docs report 313M two-second clips; evaluation on 33 tasks across six datasets.

- **Labels / granularity:** not a dataset label source; representation model.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.43 RALE Lung Sounds Repository

- **Tier:** Tier 4.
- **Source:** <https://www.rale.ca/repository.htm>.
- **Access:** web-listened education/repository; research download/metadata limited.
- **Route:** reference examples only.
- **Audio modality:** respiratory sounds.
- **Scale / format:** official page groups recordings into Normal, Wheezes, Crackles and Other; exact research-ready metadata unknown.
- **Labels / granularity:** coarse category labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.44 SNUCH Pediatric Lung Sound Dataset

- **Tier:** Tier 4.
- **Source:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC9871007/>.
- **Access:** private / not public.
- **Route:** pediatric lung-sound method evidence.
- **Audio modality:** pediatric lung sounds.
- **Scale / format:** paper reports 680 eligible training/internal-validation clips and 90 prospective-validation clips.
- **Labels / granularity:** normal/crackle/wheeze labels by pediatric pulmonologists.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.45 MIT Open Voice / cough AI preliminary resources

- **Tier:** Tier 4.
- **Source:** <https://www.media.mit.edu/projects/open-voice/overview/>.
- **Access:** project/repository status unclear.
- **Route:** speech/cough health monitoring background.
- **Audio modality:** voice/cough depending release.
- **Scale / format:** unknown / not found as stable raw-audio dataset.

- **Labels / granularity:** unknown.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.46 MEEI Voice Disorders Database

- **Tier:** Tier 4.
- **Source:** <https://masseyeandear.org/research/otolaryngology/voice-and-speech>.
- **Access:** commercial/restricted legacy dataset.
- **Route:** voice pathology background.
- **Audio modality:** voice/speech.
- **Scale / format:** secondary papers report >1,400 recordings or 657 pathological + 53 normal subsets; exact accessible package varies.
- **Labels / granularity:** voice pathology labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.47 PEEP / respiratory-support physiological datasets without raw audio

- **Tier:** Tier 4.
- **Source:** <https://physionet.org/>.
- **Access:** open/non-audio; not central.
- **Route:** clinical context only.
- **Audio modality:** not raw respiratory audio.
- **Scale / format:** varies.
- **Labels / granularity:** not acoustic labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.48 Open-source AI Stethoscope / COPD HeAR demo resources

- **Tier:** Tier 4.
- **Source:** <https://github.com/search?q=open-source+AI+stethoscope+COPD+HeAR&type=repositories>.
- **Access:** GitHub/demo; paper/dataset status not fully verified.
- **Route:** method lead only.
- **Audio modality:** stethoscope lung sounds.
- **Scale / format:** repo claims COPD AUROC but source dataset/access need verification.

- **Labels / granularity:** unknown.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.49 Kaggle ICBHI respiratory-sound mirrors

- **Tier:** Tier 4.
- **Source:** <https://www.kaggle.com/datasets/vbookshelf/respiratory-sound-database>.
- **Access:** open mirrors; often unofficial.
- **Route:** convenient download only, not primary source.
- **Audio modality:** mirrored ICBHI lung sounds.
- **Scale / format:** 920 recordings when faithful; mirror metadata may drift.
- **Labels / granularity:** inherits ICBHI labels.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.50 YouTube/Freesound cough compilation datasets

- **Tier:** Tier 4.
- **Source:** <https://freesound.org/>.
- **Access:** varies; often weakly documented.
- **Route:** negative/auxiliary only.
- **Audio modality:** web cough/sneeze/breath clips.
- **Scale / format:** unknown; many mirrors duplicate AudioSet/FSD/ESC.
- **Labels / granularity:** weak clip tags.
- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.51 Simulated/body-sound teaching libraries

- **Tier:** Tier 4.
- **Source:** <https://www.easyauscultation.com/>.
- **Access:** varies; educational license.
- **Route:** qualitative listening/reference only.
- **Audio modality:** lung/heart/respiratory examples.
- **Scale / format:** unknown / not research-ready unless metadata available.
- **Labels / granularity:** coarse teaching labels.

- **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
- **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.

6.52 General speech corpora sometimes used for health-audio pretraining

- **Tier:** Tier 4.
 - **Source:** <https://www.openslr.org/>.
 - **Access:** open but weak connection.
 - **Route:** pretraining only; not respiratory-health evidence.
 - **Audio modality:** ordinary speech.
 - **Scale / format:** large but not respiratory-specific.
 - **Labels / granularity:** transcript/speaker labels, not disease labels.
 - **Main caveat:** unknown fields should be verified from the official package before training or benchmarking. For Tier 3/4 resources, this report treats the dataset as auxiliary or low-actionability rather than core evidence.
 - **Recommended use:** use only for the route stated above; avoid converting weak web-audio labels or voice-pathology labels into respiratory disease labels.
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7. Dataset-To-Paper / Dataset-To-Method Map

7.1 Coverage check

本版 paper map 共 **56** 个条目，其中 **23** 个为 2024-2026 年近两年工作，满足“至少 40 篇 paper 且至少 10 篇近两年 paper”的目标。表中“近两年”用 ✓ 标记。这里的“paper/method map”包含数据集论文、挑战论文、使用公开数据集的方法论文、foundation model 论文，以及自采集但具有公开 repo/清晰模型定义/明确协议的论文。若数据私有但模型定义清晰，本报告将其标为 **not public-dataset comparable**，只作为方法启发或未来合作参考。

7.2 Expanded paper map

#	近两年	论文/资源	年份	数据集	任务	模型/方法族	评测/结果摘要	代码/复现性	可比性
1		A Respiratory Sound Database for the Development of Automated Classification	2019	ICBHI 2017	dataset/challenge definition	database paper and baseline challenge task	Official challenge uses respiratory-cycle classification; use as source for scale/labels	data open	yes, only under official split/metric
2		HF_Lung_V1: An open-access lung sound database with event annotations	2021	HF_Lung_V1	event detection	dataset plus baseline RNN/event detector	Event onset-offset labels for I/E/CAS/DAS; metrics paper-specific	data/code pages open	partly
3		HF_Lung_V2 and ensemble baseline for lung sound event detection	2022	HF_Lung_V2	event detection	ensemble classifier/detector	V2 train/test files reported; labels partly controlled	partial/code unknown	partly
4		HF_Tracheal_V1 tracheal sound database and CAS detection	2021	HF_Tracheal_V1	tracheal breath/CAS detection	dataset plus temporal detection baselines	About 10k tracheal 15-s recordings; label access varies	audio open; labels request	partly
5		SPRSound: Open-Access Pediatric Respiratory Sound Database	2022	SPRSound	pediatric respiratory event detection	dataset/challenge baselines	Original database reports 2,683 records, 292 participants, 9,089 events	data/challenge open	yes within challenge
6		COUGHVID: A cough audio dataset for COVID-19 research	2021	COUGHVID	cough quality and weak COVID analysis	dataset, expert cough-quality labels, baseline analysis	25k+ recordings and 2,800+ expert labels in paper	data open	limited; splits differ

#	近 两 年	论文/资源	年 份	数据集	任务	模型/方法族	评测/结果摘要	代码/复 现性	可比性
7		Coswara: A respiratory sounds and symptoms dataset for remote screening	20 23	Coswara	multimodal COVID/ symptom prediction	dataset and baseline models over cough/ breath/speech	2,635 individuals; self-report/test status labels	data open	limited
8		COVID-19 Sounds: A Large-Scale Audio Dataset	20 21	Cambridge COVID-19 Sounds	COVID/ symptom prediction	dataset and ML baselines	53,449 samples, 552 h, 36,116 participants; self-report status	DTA	partly within DTA
9		Sounds of COVID-19: exploring realistic performance and bias	20 21	Cambridge COVID-19 Sounds	COVID audio prediction and bias analysis	ML baselines plus bias/control analyses	Shows that confounding and bias affect apparent audio performance	code/ data controlled	not comparable to challenge metrics
10		DiCOVA Challenge: Diagnosing COVID-19 using acoustics	20 21	DiCOVA/ Coswara subset	COVID detection	challenge baselines; cough/ breath/speech tracks	Track-specific AUC/UAR metrics; official folds	challenge access	yes within track
11		Second DiCOVA Challenge baseline and protocol	20 21	DiCOVA Track 2	COVID detection	challenge baselines with expanded modalities	Subject-level binary labels and official challenge protocol	challenge access	yes within track
12		INTERSPEECH ComParE 2021 COVID-19 Cough/Speech Sub-Challenges	20 21	ComParE COVID Cough/Speech	COVID detection	challenge baselines over cough/speech features	Challenge table gives train/dev/test splits and official metrics	challenge access	yes within track
13	✓	UK COVID-19 Vocal Audio Dataset	20 24	UK COVID-19 Vocal Audio	PCR-linked disease dataset	dataset paper and baseline code	72,999 participants; 70,794 PCR-linked; open cough/exhalation release	data+code open	yes if official code/protocol used
14	✓	Audio-based AI classifiers show no evidence of improved COVID-19 screening over symptom checkers	20 24	UK COVID Vocal; COVID-19 Sounds	disease prediction validity	comparative baselines against symptoms	Important negative result: audio classifiers did not show useful improvement over symptom checkers in tested settings	code/ data links available via project	partly; focus is validity
15		Sound-Dr: A dataset and benchmark for COVID-19 and respiratory symptoms	20 23	Sound-Dr	COVID/ symptom/ anomaly prediction	CNN/RNN/ML baselines over cough and breathing	3,930 recordings, 1,310 subjects; paper-specific metrics	GitHub open	partly
16		smarty4covid: A COVID-19 audio knowledge base	20 23	smarty4covid	cough/ breath/ voice dataset and baseline tasks	dataset plus modality/segment baselines	18,265 recordings, 4,673 users	data release described	limited

#	近 两 年	论文/资源	年 份	数据集	任务	模型/方法族	评测/结果摘要	代码/复 现性	可比性
17	✓	A deep learning framework to detect COVID-19 from cough audio signals	2025	COUGHVID	COVID cough segmentation/ classification	U-Net style cough segmentation plus deep classifier	Paper reports COVID cough prediction pipeline; metrics source-specific	code unknown	limited
18	✓	SympCoughNet: symptom-aware cough network for COVID-19 audio	2025	COVID cough datasets	COVID cough+symptom prediction	dual-channel cough and symptom weighting/fusion network	Reported results are dataset/protocol-specific	code unknown	limited
19		Project Achoo: a practical cough-based COVID-19 detection app	2022	self-collected / app data	COVID cough app	ML pipeline with mobile app data	Paper-specific validation; useful engineering reference	code unknown	not comparable
20		COVID-19 detection using cough, breath and speech with deep learning	2021	Coswara and related respiratory audio	multimodal COVID prediction	CNN/RNN/feature-fusion models	Metrics depend on dataset filtering and self-report labels	code unknown	limited
21		A C-BiLSTM cough event boundary detector and Corp Dataset	2022	Corp Dataset	long-recording cough event detection	C-BiLSTM boundary regression/ detection	Source reports sensitivity 84.13%, specificity 99.82%, IoU 0.89	data page linked; code unknown	partly; patient split required
22		Automated cough detection in a clinic waiting room	2021	Clinic Waiting Room Cough	cough event detection	YAMNet-style/ ensemble cough detector	Paper reports high ROC-AUC around 98 in validation and deployment testing	project URL; raw data unclear	partly
23	✓	Audio-based cough event detection on multimodal wearable audio-IMU data	2024	Audio-IMU Cough Dataset	cough event detection	audio, IMU and multimodal classifiers	Dryad abstract reports 92.59% in-subject and 90.79% cross-subject accuracy	data open	partly
24		VocalSound: A dataset for human non-speech vocalization recognition	2022	VocalSound	vocal event recognition	CNN/Transformer-style baselines for cough/sneeze/ sniff etc.	21,024 recordings, 3,365 subjects; six classes	data+code available	yes for vocal-event task
25		Hybrid CNN-LSTM lung-sound classification with focal loss	2022	ICBHI 2017	lung sound classification	hybrid CNN-LSTM / focal-loss variants	Uses multiple splitting schemes; exact results depend on split	code unknown	only if split matched
26		Patch-Mix Contrastive Learning with Audio Spectrogram Transformer	2023	ICBHI 2017	respiratory sound classification	AST plus patch-level mix and contrastive learning	GitHub reports ICBHI experiments; metrics split-dependent	code open	partly
27		Pretraining Respiratory Sound Representations Using Metadata and Contrastive Learning	2023	ICBHI; SPRSound	representation learning	metadata-aware contrastive pretraining	Official PyTorch includes ICBHI/SPRSound workflows	code open	partly

#	近 两 年	论文/资源	年 份	数据集	任务	模型/方法族	评测/结果摘要	代码/复 现性	可比性
28	✓	Stethoscope-Guided Supervised Contrastive Learning for Cross-Domain Adaptation	2024	ICBHI 2017	lung sound classification/domain adaptation	supervised contrastive learning plus stethoscope/domain adversarial training	Repo/source reports ICBHI score 61.71 and +2.16 over baseline	code open	yes if official split
29	✓	Multi-View Spectrogram Transformer for Respiratory Sound Classification	2024	ICBHI 2017	lung sound classification	multi-view mel spectrogram transformer	ICBHI experiments; reports SOTA-family score in 2024 papers	code open	yes if official split
30	✓	BTS: Metadata-Aided Respiratory Sound Classification	2024	ICBHI 2017	lung sound classification	text-audio / metadata-aided model using stethoscope/location metadata	Public sources report strong ICBHI score around 63.54; verify exact table before citation	code unknown/limited	yes if official split
31	✓	RepAugment: Input-Agnostic Representation-Level Augmentation	2024	ICBHI 2017	lung sound classification	representation-level augmentation on pretrained respiratory/audio models	Paper reports minority-class improvements and ICBHI gains	code open	yes if official split
32	✓	OPERA: Towards Open Respiratory Acoustic Foundation Models	2024	OPERA benchmark datasets	foundation model benchmark	respiratory acoustic pretraining and 19 downstream tasks	OpenReview/GitHub report improvement over general acoustic models on 16/19 tasks	code open	benchmark-level, not per-dataset universal
33	✓	HeAR: Health Acoustic Representations	2024	multiple health-audio datasets	foundation model	large-scale ViT-MAE-style health acoustic embeddings	Paper/model docs report 313M 2-s clips and evaluation on 33 tasks across six datasets	model gated/open docs	task-dependent
34	✓	JMIR AI FABS audio enhancement / breath-sound classification paper	2025	FABS; ICBHI; NTUH noise	lung abnormality classification under noise	deep learning with physician-annotated FABS and noise robustness experiments	Scale: FABS 5,238 recordings; results paper-specific	code/data unknown	not yet comparable
35	✓	Patient-Aware Feature Alignment for Robust Lung Sound Classification	2025	ICBHI 2017	lung sound classification	patient-aware feature alignment / robustness method	Public sources report SOTA-family ICBHI results; exact comparability needs split audit	code unknown/limited	yes if official split
36	✓	Architecture-Agnostic Knowledge Distillation from Ensembles	2025	ICBHI 2017	lung sound classification	ensemble teacher to lightweight/student models	Interspeech/GitHub release; results compare with BTS/PAFA/M2D in paper	code open	yes if official split
37	✓	Adaptive Differential Denoising for Respiratory Sounds Classification	2025	ICBHI 2017	lung sound classification/denoising	adaptive differential denoising with respiratory classifier	Sources report ICBHI Score 65.53 and +1.99 over previous SOTA	code open	yes if official split

#	近 两 年	论文/资源	年 份	数据集	任务	模型/方法族	评测/结果摘要	代码/复 现性	可比性
38	✓	Towards Pre-training an Effective Respiratory Audio Foundation Model	2025	OPERA; ICBHI; HF_Lung; COUGHVID; AudioSet	foundation / pretraining comparison	M2D+Resp and comparison of 21 audio foundation models	Paper reports M2D+Resp average AUROC 0.814 vs prior OPERA-CT 0.733 over OPERA tasks	code open/ source public	benchmark-level
39	✓	Geometry-Aware Optimization for Robust Respiratory Sound Classification	2025	ICBHI 2017	lung sound classification	AST with sharpness/ geometry-aware optimization	Public sources report official ICBHI score 68.10 and sensitivity 68.31; verify arXiv/repo version	code open	yes if official split
40	✓	Resp-Agent: Multimodal Respiratory Sound Generation and Disease Diagnosis	2026	Resp-229K + downstream datasets	multimodal respiratory diagnosis/ generation	agentic curriculum, modality-weaving diagnoser, flow-matching generator	Introduces 229k-recording/407+h Resp-229K; claims >5 point official ICBHI improvement over prior method	code+data announced	needs independent audit
41	✓	E-RespiNet: LLM-ELECTRA driven triple-stream CNN	2025	ICBHI and/or respiratory sound datasets	respiratory sound classification	LLM-ELECTRA/ triple-stream CNN design	PMC paper reports high respiratory sound classification results; protocol must be checked	code unknown	only if split matched
42	✓	Enhanced respiratory sound classification using deep noise suppression	2025	ICBHI/ respiratory sound datasets	lung sound classification	deep denoising plus classifier pipeline	JCM/PMC paper-specific results; verify split before comparison	code unknown	limited
43	✓	Pediatric Asthma Detection with Google HeAR	2025	SPRSound	pediatric asthma-indicative modeling	HeAR embeddings plus downstream classifier	Preprint/source reports over 91% accuracy; label/ task semantics require audit	code unknown	limited
44	✓	iMedic: smartphone self-auscultation for pediatric pneumonia/ abnormal respiratory sounds	2025	self-collected smartphone dataset	pediatric abnormal respiratory sound screening	mobile self-auscultation ML pipeline	Model/protocol described; raw data access not confirmed	code unknown	not comparable
45		DeepBreath: Deep Learning of Breathing Patterns for Automatic Pneumonia Diagnosis	2023	DeepBreath/ Pneumoscope private dataset	pediatric diagnosis/ pathology	CNN/attention architecture with internal/external validation	Repo reports AUROC 0.93/0.89 healthy-vs-pathologic internal/ external; other disease AUROCs reported	code open; raw data request	not public-dataset comparable
46		Deep learning classification of pediatric lung sounds from SNUCH	2023	SNUCH private pediatric lung sounds	normal/ crackle/ wheeze classification	classical ML and CNN-style classifiers	Paper reports prospective validation; exact metrics task-dependent	code/ data unknown	not comparable
47	✓	Neural Cough Counter	2024	self-collected / field cough data	cough counting/ event detection	neural cough event detector for long monitoring	Paper model definition is clear; raw data/code status requires check	code unknown	not comparable unless release found
48		VOICED: A validated database for voice disorder research	2018	VOICED	voice pathology	dataset resource and baseline voice pathology work	208 samples; 150 pathological and 58 healthy	data open PhysioNet	voice-task only

#	近 两 年	论文/资源	年 份	数据集	任务	模型/方法族	评测/结果摘要	代码/复 现性	可比性
4 9		Saarbruecken Voice Database common protocol paper	20 21	SVD	voice pathology	common-protocol benchmark and critique of prior non-comparability	Paper emphasizes that previous SVD results are often non-comparable without common splits	data open/site	voice-task only
5 0		ComParE 2020 Breathing Sub-Challenge	20 20	UCL Speech Breath Monitoring	breathing-from-speech regression	audio features and regression baselines to respiratory belt signal	49 speakers, 4-min spontaneous speech each; official train/dev/test	challenge data	yes within challenge
5 1		ESC-50: Dataset for Environmental Sound Classification	20 15	ESC-50	auxiliary audio pretraining	dataset/baselines; includes cough/breath/sneeze/snore classes	2,000 5-s clips, 50 classes, 5 folds	data open CC BY-NC	generic only
5 2		FSD50K: an open dataset of human-labeled sound events	20 22	FSD50K	auxiliary audio pretraining	large Freesound weak-label dataset	51,197 clips, 200 classes, >100 h	data open with licenses	generic only
5 3		AudioSet: An ontology and human-labeled dataset for audio events	20 17	AudioSet	generic pretraining	weakly labeled YouTube audio ontology	Includes cough/breath/sneeze/snore-related labels; Cough class page reports 871 clips	metadata open	generic only
5 4		cough-speech-sneeze nonverbal corpus paper/release	20 17	cough-speech-sneeze	auxiliary cough/sneeze/speech detection	nonverbal audio event classification	YouTube-derived cough/sneeze/speech/silence corpus	audb/audformat tooling	generic only
5 5		OpenSLR Nonverbal Vocalization Dataset	20 21	OpenSLR NVD	auxiliary nonverbal vocalization recognition	16-class nonverbal vocalization corpus	Includes cough, yawn, throat-clearing, sigh, panting, sneeze etc.	data open	generic only
5 6		Class-imbalanced voice pathology detection and database critique	20 21	MEEI/SVD voice datasets	voice pathology	class-imbalance-aware voice pathology models	Useful for speech-health evaluation risks, not respiratory acoustic events	code unknown	voice-task only

7.3 Current SOTA snapshot and comparability notes

ICBHI 2017 是当前最集中、最容易出现“看似 SOTA 但不可比”的数据集。应只把使用官方 split、官方 score 或明确 patient-independent protocol 的结果放在同一表中比较。2024-2026 年可重点复现或对照的方向包括 MVST、BTS、RepAugment、Stethoscope-guided SupCon、ADD-RSC、Architecture-

Agnostic KD、Geometry-aware AST+SAM 和 Resp-Agent。公开来源中，ADD-RSC 报告 ICBHI Score 65.53，Geometry-aware AST+SAM 报告 Score 68.10，Resp-Agent 报告进一步提升；这些结果需要逐条核查 split、augmentation、pretraining 数据是否包含测试污染、是否使用 patient metadata，以及是否和官方 ICBHI score 完全同口径。

OPERA / respiratory foundation models 的近两年趋势是从 generic AudioSet-pretrained model 转向 health/respiratory-specific representation。OPERA 2024 提供 19-task benchmark；Niizumi et al. 2025 在 OPERA route 上比较 21 个 audio foundation models，并报告 M2D+Resp 的平均 AUROC 相比 OPERA-CT 明显提升。这个方向适合写成 initial result: frozen embedding + lightweight classifier + cross-dataset transfer，而不是从零训练大型模型。

COVID/cough disease prediction 的最新重点不是单纯追求 AUC，而是验证 audio 是否相对 symptom/metadata baseline 有增量。UK COVID Vocal 2024 和 Coppock et al. 2024 的结果非常重要：任何 COVID audio paper 都应加入 symptom-only、metadata-only 和 audio+symptom 对比，避免把 recruitment/device/symptom confounds 当作声学疾病信号。

自采集但可复现论文 对 long-term collaboration 很有价值。DeepBreath、FABS、SNUCH、Clinic Waiting Room Cough、Audio-IMU Cough、Neural Cough Counter、iMedic 等代表真实采集路线。短期不一定能下载原始数据，但可以复现模型定义、标注流程、site-independent validation 和 deployment metrics 的写法，为 UNC Hospital 后续真实数据采集设计提供模板。

8. Cross-Dataset Aggregation And Compatibility Analysis

8.1 Label compatibility

跨数据集合并时至少要拆成五层标签：

1. **Acoustic event layer:** normal, crackle/fine crackle/coarse crackle, wheeze, rhonchi, stridor, cough, breath/inhalation/exhalation, poor quality/noise。
2. **Respiratory phase layer:** inhalation, exhalation, pause/hold, respiratory belt trajectory。
3. **Symptom layer:** cough symptom, dyspnea, fever, sore throat, nasal congestion, self-reported wheeze。
4. **Disease/test layer:** PCR-linked COVID, self-reported COVID, antigen/test status, TB microbiology, COPD/asthma/pneumonia/bronchiolitis diagnosis, voice pathology。
5. **Confound/metadata layer:** age, sex/gender, language, country, device, stethoscope, chest location, hospital/site, recording date, recruitment route, microphone distance, recording environment。

可以防守的映射包括：ICBHI crackle/wheeze、SPRSound crackle/wheeze/rhonchi/stridor、HF CAS/DAS 到粗粒度 adventitious sound layer；COUGHVID/VocalSound/AudioSet Cough 到 cough event layer；UK Vocal/Cambridge/Coswara 到 disease/test layer但保留 PCR-linked vs self-report 区分。不可防守的映射包括：把 self-reported COVID positive 等同于 PCR positive；把 wheeze event 等同于 asthma diagnosis；把 voice pathology 等同于 respiratory disease。

8.2 Audio/device compatibility

Body-contact lung sounds、tracheal sounds、smartphone cough、far-field clinic audio、web speech 和 generic YouTube audio 的传递函数完全不同。HF_Lung_V1 的 4 kHz/16-bit body-sound data 与 COUGHVID 的 web audio、UK Vocal 的 WAV cough/exhalation、AudioSet 的 YouTube weak labels 不应在同一个 front-end 下默认为同分布。跨数据集实验应报告 resampling、normalization、band-pass filtering、silence trimming、quality filtering 和 device/site metadata use。

8.3 Split and leakage rules

最低协议要求是 **subject-independent split**。对长录音或 repeated recording 数据，还需要 session-independent 或 site-independent split。特别危险的泄漏包括：ICBHI cycle-level random split；HF/KAUH/SPRSound 的 same patient multiple files；COUGHVID repeated users/devices；Corp/Clinic waiting-room 的同一 session 切片；Resp-229K 与下游测试集源重叠；foundation model 在 evaluation data 上的 pretraining contamination。

8.4 Pediatric/adult mismatch

SPRSound、DeepBreath、SNUCH、Bridge2AI Pediatric 属于 pediatric route。儿童呼吸音的 airway size、疾病谱、哭声/说话噪声、合作程度、设备位置和临床标签与成人数据明显不同。应把 pediatric route 独立报告，或作为 cross-age adaptation/transfer learning，而不是和成人 ICBHI/HF/KAUH 无条件合并。

8.5 Clinical-label validity

疾病预测论文的核心不是“模型能不能从音频预测标签”，而是“标签是否可靠、预测是否相对症状/元数据有增量、结果是否在外部 cohort 上保持”。UK Vocal 强于 self-report COVID datasets，因为其标签与 PCR 结果连接；但即便 PCR-linked，也仍需处理 test date、症状状态、设备、人口统计和 recruitment confounds。

9. Label-Semantics Risks And Protocol Risks

1. **Disease label vs acoustic-event label:** wheeze/crackle/cough 是声学事件，不是疾病诊断。
2. **Self-report vs PCR-linked label:** Coswara/Cambridge/COUGHVID/Sound-Dr/smarty4covid 多包含自报告或混合状态；UK Vocal 的 PCR-linked 标签更强，但也不能消除采样偏差。
3. **Symptom label vs disease label:** cough、dyspnea、fever、wheezing symptom 是症状，不能当作病种标签。
4. **File-level vs onset-offset labels:** file-level abnormality 不能直接和 onset-offset event detection 比较。
5. **Patient/session leakage:** 随机切 cycle/clip/event 会显著虚高结果。
6. **Weak web labels:** AudioSet/FSD/ESC/OpenSLR 标签不是临床标签。
7. **Foundation-model contamination:** 大规模 web/health pretraining 可能已经见过公开测试音频或其近似版本；需要 reporting。
8. **Derived-label risk:** Resp-229K 的 LLM-distilled narrative 对方法有启发，但标签语义必须审计。
9. **Access reproducibility:** FABS/DeepBreath/SNUCH/Cambridge/DiCOVA/ComParE 受控或私有，不能作为短期 guarantee。

10. **Metadata shortcuts:** device/site/country/date/symptom can dominate; disease-prediction models must include metadata-only baselines.

10. Project-Fit Recommendations

10.1 最适合短期 **initial result** 的路线

建议短期做 **frozen foundation/audio representation + lightweight classifier + robust split audit**，而不是从零训练复杂模型。可组合三条实验：

- **Lung event route:** ICBHI official split + HF_Lung_V1 event-level + SPRSound pediatric external validation。模型包括 CNN/AST/MVST/BTS-style metadata variant/HeAR/OPERA/M2D embeddings。
- **Cough event route:** COUGHVID cough-quality + Audio-IMU cough + Clinic/Corp paper protocol reference。重点是 cough/non-cough、quality filtering、boundary detection，不直接疾病诊断。
- **Disease-label route:** UK COVID Vocal audio-only vs symptom-only vs audio+symptom。把 Coppock et al. 2024 作为必须讨论的 negative benchmark。

10.2 最适合投 **workshop / short paper** 的路线

一个可控题目是：**Do respiratory acoustic foundation embeddings transfer across lung-sound, cough, and pediatric respiratory event datasets?** 具体实验是统一前处理、统一 subject-independent split、比较 MFCC/CNN、AST/BEATs、HeAR、OPERA/M2D-style embeddings，在 ICBHI/HF/SPRSound/COUGHVID/Audio-IMU 上报告同口径指标。贡献点不是“临床诊断”，而是 dataset landscape-aware benchmark 和 protocol audit。

10.3 长期 UNC Hospital collaboration 路线

长期应转向 **prospective data collection + clinician annotation + external validation**。DeepBreath、FABS、SNUCH、Clinic Waiting Room Cough 和 iMedic 的价值是提供采集/标注/验证设计模板：记录设备、位置、环境噪声、症状、诊断、年龄、性别、临床路径；确保 train/test 按 patient/site/time 分离；同时保留声学事件标签和疾病/症状标签两层。

10.4 Speech-health / telehealth route

Bridge2AI-Voice Adult/Pediatric、VOICED、SVD、MEEI 等适合做 voice health / speech-health background，而不是 lung-sound 替代。可发展为远程健康监测路线：speech-breath coupling、voice pathology、respiratory symptom self-report plus acoustic biomarkers、multimodal clinical metadata。短期不要把这条线和 stethoscope lung-sound event detection 混成一个 label space。

11. Open Questions And Access Blockers

1. **Cambridge COVID-19 Sounds:** DTA 获取流程、当前可用版本、是否包含全部 cough/breath/voice。
2. **DiCOVA/ComParE data:** challenge 数据是否仍可申请，是否允许新论文复现。
3. **Corp Dataset:** 原始长录音和标注是否可下载，是否有 DTA。
4. **FABS:** 是否计划公开，或只能通过合作访问。
5. **DeepBreath/Pneumoscope:** reasonable request 的实际门槛、IRB/consent 限制和可用字段。

6. **Resp-229K**: 需要审计源数据重叠、license compatibility、LLM narrative 生成方式和 downstream test contamination。
7. **Bridge2AI raw audio**: PhysioNet release 主要是 derived features；raw audio 需要 Synapse/DACO 和机构签署。
8. **HF_Lung_V2 / HF_Tracheal labels**: 部分标签访问路径需要表单/请求。
9. **SOTA comparability**: ICBHI 新论文的 score 必须逐个核查 split、pretraining、augmentation 和 leakage。
10. **Licenses**: COUGHVID/UK Vocal/HF/AudioSet/FSD/VocalSound/ESC 的许可证不一致，不能直接重发布融合数据集。

12. Source Bibliography With URLs

#	Source	URL
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2	ICBHI challenge page	https://bhichallenge.med.auth.gr/ICBHI_2017_Challenge
3	Rocha et al. ICBHI database PubMed	https://pubmed.ncbi.nlm.nih.gov/30708353/
4	HF_Lung_V1 GitLab	https://gitlab.com/techsupportHF/HF_Lung_V1
5	HF_Lung_V1 PLOS ONE	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0254134
6	HF_Lung_V2 Applied Sciences	https://www.mdpi.com/2076-3417/12/15/7623
7	HF_Tracheal_V1 GitLab	https://gitlab.com/techsupportHF/HF_Tracheal_V1
8	SPRSound GitHub	https://github.com/SJTU-YONGFU-RESEARCH-GRP/SPRSound
9	SPRSound paper DOI	https://doi.org/10.1109/TBCAS.2022.3204910
10	COUGHVID EPFL	https://www.epfl.ch/labs/esl/index-html/datasets/coughviddataset/
11	COUGHVID Zenodo	https://zenodo.org/records/7024894
12	COUGHVID Scientific Data	https://www.nature.com/articles/s41597-021-00937-4
13	UK COVID Vocal Zenodo	https://zenodo.org/records/10043978
14	UK COVID Vocal GitHub	https://github.com/alan-turing-institute/Turing-RSS-Health-Data-Lab-Biomedical-Acoustic-Markers
15	UK COVID Vocal Scientific Data	https://www.nature.com/articles/s41597-024-03132-3
16	Coppock et al. audio vs symptoms	https://www.nature.com/articles/s42256-024-00815-3
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1 9	RespiratoryDatabase@TR Mendeley	https://data.mendeley.com/datasets/p9z4h98s6j/1
2 0	Coswara GitHub	https://github.com/iiscleap/Coswara-Data
2 1	Coswara arXiv	https://arxiv.org/abs/2305.12741
2 2	Cambridge COVID-19 Sounds OpenReview	https://openreview.net/forum?id=9KArJb4r5ZQ
2 3	COVID-19 Sounds DTA blog	https://www.covid-19-sounds.org/en/blog/neurips_dataset
2 4	Sounds of COVID-19 npj Digital Medicine	https://www.nature.com/articles/s41746-021-00553-x
2 5	DiCOVA challenge	https://dicovachallenge.github.io/
2 6	DiCOVA Interspeech paper	https://www.isca-archive.org/interspeech_2021/muguli21_interspeech.pdf
2 7	ComParE challenge summary	https://pmc.ncbi.nlm.nih.gov/articles/PMC10031089/
2 8	Sound-Dr GitHub	https://github.com/ReML-AI/Sound-Dr
2 9	Sound-Dr paper	https://papers.phmsociety.org/index.php/phmap/article/download/3604/2088
3 0	smarty4covid Scientific Data	https://www.nature.com/articles/s41597-023-02646-6
3 1	Virufy data GitHub	https://github.com/virufy/virufy-data
3 2	Audio-IMU Cough Dryad	https://datadryad.org/dataset/doi:10.5061/dryad.0p2ngf26t
3 3	VocalSound official	https://groups.csail.mit.edu/sls/downloads/vocalsound/
3 4	VocalSound GitHub	https://github.com/YuanGongND/vocalsound
3 5	Pulmonary Sound Mendeley	https://data.mendeley.com/datasets/6mnp9v3n73/1
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3 7	DeepBreath GitHub	https://github.com/pneumonia-diagnosis/DeepBreath
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4 2	HLS-CMDS Mendeley	https://data.mendeley.com/datasets/pxg5f7j5kd/1

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4 4	AudioSet ontology	https://research.google.com/audioset/ontology/index.html
4 5	AudioSet Cough class	https://research.google.com/audioset/ontology/cough_1.html
4 6	FSD50K Zenodo	https://zenodo.org/records/4060432
4 7	ESC-50 GitHub	https://github.com/karolpiczak/ESC-50
4 8	cough-speech-sneeze docs	https://audeering.github.io/datasets/datasets/cough-speech-sneeze.html
4 9	OpenSLR Nonverbal Vocalization	https://www.openslr.org/99/
5 0	FluSense ACM DL	https://dl.acm.org/doi/10.1145/3381014
5 1	VOICED PhysioNet	https://physionet.org/content/voiced/
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5 3	Bridge2AI Adult PhysioNet	https://physionet.org/content/b2ai-voice/
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5 5	Bridge2AI Voice project	https://bridge2ai.org/data-voice/
5 6	OPERA GitHub	https://github.com/evelyn0414/OPERA
5 7	OPERA website	https://evelyn0414.github.io/OPERA/
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5 9	Niizumi et al. 2025 respiratory pretraining	https://www.isca-archive.org/interspeech_2025/niizumi25_interspeech.html
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6 1	HeAR arXiv	https://arxiv.org/abs/2403.02522
6 2	MVST GitHub	https://github.com/xingmingyu123456/MVST
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6 4	Stethoscope-guided SupCon GitHub	https://github.com/ilyassmoummad/stethoscope-guided-supervised-contrastive-learning
6 5	RepAugment GitHub	https://github.com/june-oh/RepAugment
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71	E-RespiNet PMC	https://pmc.ncbi.nlm.nih.gov/articles/PMC11938052/
72	SNUCH pediatric lung-sound paper	https://pmc.ncbi.nlm.nih.gov/articles/PMC9871007/
73	RALE Repository	https://www.rale.ca/repository.htm
74	MEEI Voice and Speech Research	https://masseyeandear.org/research/otolaryngology/voice-and-speech

Final Summary

- **Total number of datasets/resources covered:** 52.
- **Datasets per relevance tier:** Tier 1 = 6; Tier 2 = 20; Tier 3 = 16; Tier 4 = 10.
- **Paper/method map size:** 56 entries, including 23 entries from 2024-2026.
- **Strongest directly useful datasets:** ICBHI, HF_Lung_V1, SPRSound, COUGHVID, UK COVID Vocal, Corp Dataset. They cover lung event labels, pediatric respiratory events, cough quality/event detection, PCR-linked COVID audio, and long-duration cough boundaries.
- **Important datasets with major issues:** Cambridge/DiCOVA/ComParE are access/challenge constrained; Coswara/COUGHVID/Sound-Dr/smarty4covid have self-report or mixed labels; FABS/DeepBreath/SNUCH are clinically relevant but raw data are not fully open; Resp-229K is powerful but derived/LLM-labeled and needs contamination audit; AudioSet/FSD/ESC/OpenSLR are non-clinical weak-label resources.

- **Best short-term research route:** respiratory event detection and cross-dataset robustness with ICBHI + HF_Lung + SPRSound, plus cough-quality/event detection using COUGHVID and Audio-IMU Cough. Add HeAR/OPERA/M2D-style frozen embeddings as baseline.
- **Best disease-label route:** UK COVID Vocal with symptom-only/audio-only/audio+symptom comparison. Do not claim clinical screening value without external validation.
- **Best long-term telehealth route:** prospective clinical audio collection with two-layer labels: acoustic events plus clinical/symptom outcomes; use DeepBreath/FABS/Clinic Waiting Room/iMedic as study-design references and Bridge2AI-Voice as speech-health context.
- **Biggest unresolved questions:** controlled dataset access, patient-independent splits, cross-source label mapping, license compatibility, foundation-model contamination, and whether audio provides incremental value beyond symptoms/metadata.
- **PDF generation status:** PDF 已生成，并完成抽查渲染；使用中文字体和横向 A4 表格布局。